



Exercise Session 1

📅 22.09.2025

Algorithms & Data Structures HS 25 G-21

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Plan for Today

1. Organization

- About me
- WhatsApp group
- Course Structure
- Working Groups
- Exercise Session Organization
- Exam Information

2. Theory recap

- Induction

3. In Cass Exercises

- Exercise Sheet 0
- Additional Exercises
- Quiz preparation

Organization > About me / WhatsApp group

- Hi, everyone! My name is Neli Tsokanova.
- I am in third semester BSc CS at ETH Zurich.
- My ETH email is: ntsokanova@ethz.ch
- Join the WhatsApp group for this exercise class.

Organization > Course Structure

✓ Main topics covered by this course

1.	Asymptotic Notation
2.	Searching and Sorting Algorithms
3.	Data Structures
4.	Dynamic Programming
5.	Graphs

✓ Homework and Bonus Points

- Moodle Quizzes
- Code Expert Programming Exercises
- Theory Exercises
- Peer Grading
- Bonus Points Formula

✓ Moodle Quizzes

- Done individually at the beginning of each exercise class.
- Each quiz typically contains 5 exercises and has duration 5 minutes.
- The quiz is password protected and can only be taken between 9:15 and 9:30. I will give you the quiz password at the beginning of the session and you can start the quiz.
- If you start the quiz later than 9:25 you will NOT have the full 5 minutes to complete the quiz! You will only have the remaining time until 9:30!
- The quizzes are available both in German and English depending on which language you choose from the settings on Moodle.
- We recommend that you use Moodle ONLINE instead of the app version for the quiz!
- Bring up to 1 bonus point
- The quizzes will be discussed in class.
- First Quiz next Monday (29.09.2025).

✓ Code Expert Exercises

- Done individually in Java.
- For each set of Code Expert Exercises you will have 2 weeks to complete them.
- Bring up to 6 bonus points.
- No additional imports are allowed other than imports already included in the code template!
- The Code Expert Exercises will be discussed in class.

✓ Theory Exercises

- Done in working groups.
- The exercise sheets are published on Monday and your solutions have to be submitted on Moodle by 23:59 o'clock on Sunday the same week, otherwise you won't be able to submit them.
- Only one person of the group needs to submit the solutions of the group.
- All members of the group receive the same number of points for the exercises.
- In order to receive the full amount of bonus points for the exercise sheet, it is enough to submit correct and well written solutions to only the graded exercises. If you want feedback on other exercises from the exercise sheet, submit them and they will be corrected as well.
- Bring up to 3 bonus points.
- Theory exercises will be discussed in class.

✓ Peer Grading

- Done in working groups
- Each working group receives the solutions of another working group and needs to give corrections to the solution of one of the graded exercises of the Exercise Sheet.
- You receive the solutions you need to grade on Monday and you need to submit your corrections on Moodle by 23:59 on the same day.
- Motivation:
 1. By reading your peers' solutions you gain a broader perspective how the problems can be approached.
 2. By correcting others solutions you notice what makes a solution unclear or easy to follow so that you can improve your own work.
- Brings up to 1 bonus point.

✓ Bonus Points

- The main purpose of the bonus points is to give you motivation and to provide guidance to you about what to work on during the semester.
- Don't stress too much if you loose a bonus point 😊.
- Bonus calculation formula:

$$\text{Final Bonus} \begin{cases} 0,25 & \text{if } n_points \geq 0.8 * \text{max_points} \\ 0,25 (n_points / (0.8 * \text{max_points})), & \text{otherwise} \end{cases}$$

✓ Tips from my experience with the course

- Consistency and self preparation are important.
- Don't rely only on the lectures and the exercise classes. You need to spend some time going over the material and solving the exercises by yourself regularly during the semester.
- Give yourself time to relax and have fun.

Organization > Working groups

- Every 3 weeks you will be divided into working groups of 2 or 3 people and you will need to solve the Theory Exercises and complete the peer Grading within these groups.
- Motivation: Working into groups provides you with experience in collaborating with others and allows you to see different ideas related to the solutions of the exercises.

Organization > Exercise Organization / Exam Information

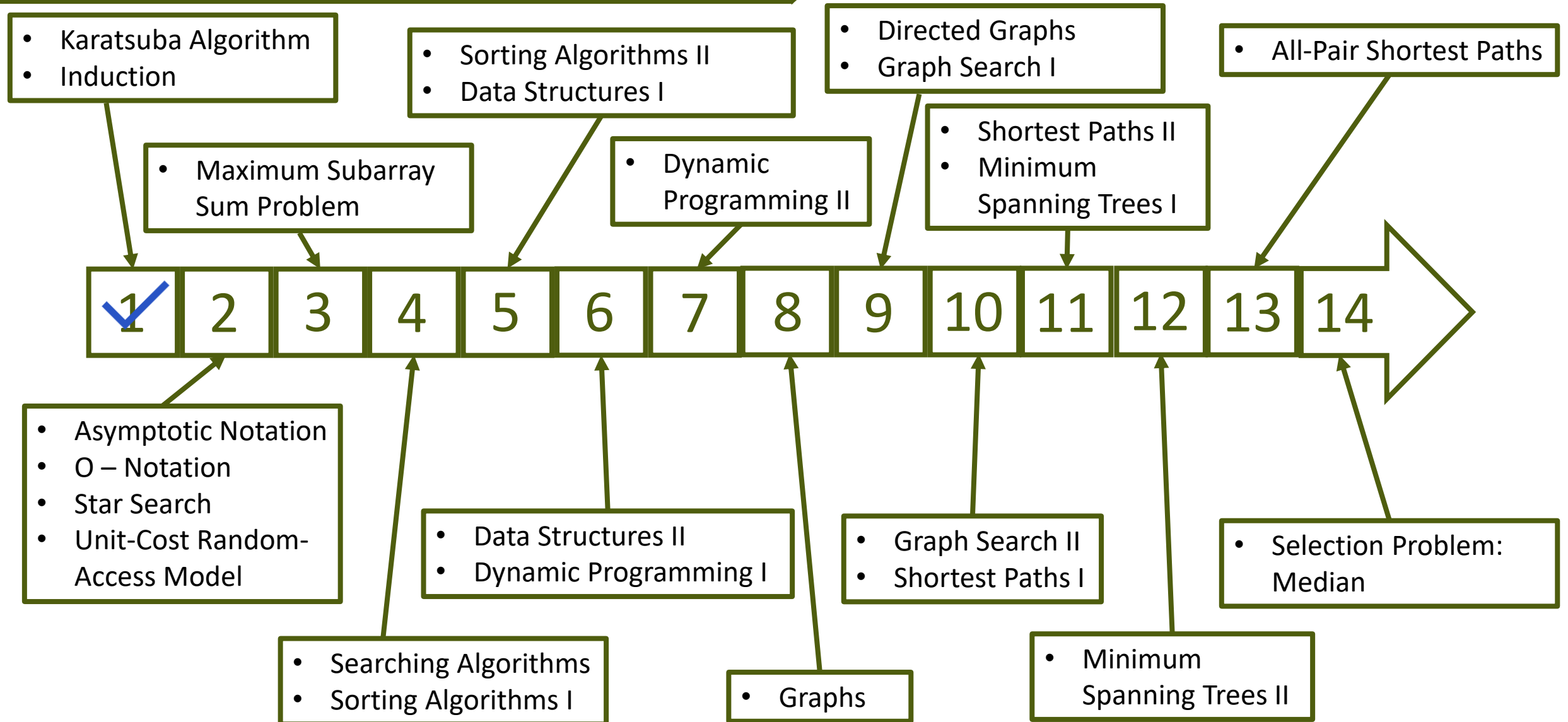
✓ Exercise Session Organization

1. Moodle Quiz + Discussing the exercises from the quiz
2. Common mistakes from the exercise sheet from two weeks ago
3. Theory exercises from the previous week
4. Code expert exercises from the previous two weeks (every other week)
5. Theory recap from the lecture
6. In class exercises
7. Preparation for next weeks Moodle Quiz

✓ Exam Information

- Change in the exam format
- 4 hours both theory and coding exercises
- Part of the exam will be on paper and the rest will be on computer
- There will be a mock exam following the format of the final exam

Course Timeline



Theory Recap > Induction

Mathematical induction

Def. Let $A(k)$ be a statement for $k \in \mathbb{N}$. Mathematical induction is a proof method used to show that a given statement (for example A) is true for all natural numbers* (i.e. $A(n)$ holds for $\forall n \in \mathbb{N}$). It consists of 2 steps:

1. **Base Case:** Prove that the statement is true for the first number(s)* (for example show that $A(1)$ holds)

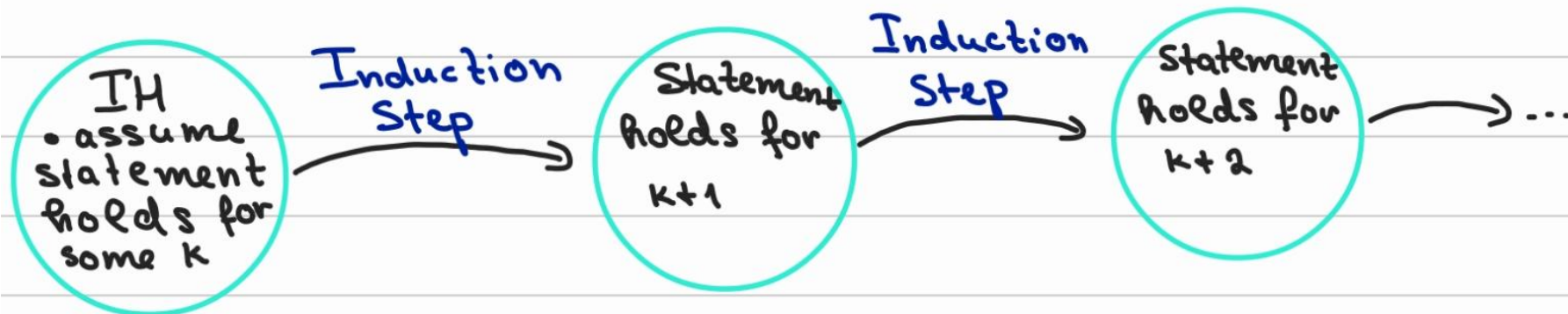
2. **Induction step:** Here we need to show that for every number $n \in \mathbb{N}$, that if the statement is true for n , then it is also true for $n+1$ (i.e. if $A(n)$ holds, then $A(n+1)$ also holds).

In order to do that, we assume that the statement holds for some arbitrary $k \in \mathbb{N}$ (**Induction Hypothesis**) and using that assumption we prove that the statement also holds for $k+1$

How do the Base Case, Induction Hypothesis and Induction Step fit together?

In the Induction Hypothesis we assume that the statement holds for some $k \in \mathbb{N}$. Using this assumption in the Induction Step we show that the statement holds for $k+1$. Since we have chosen k arbitrarily and we have shown that if the statement holds for k , then it holds for $k+1$

$\Rightarrow$ since we know that the statement holds for $k+1$ it automatically follows that the statement holds for $k+2$ and so on. This allows us to show that the statement holds for all natural numbers

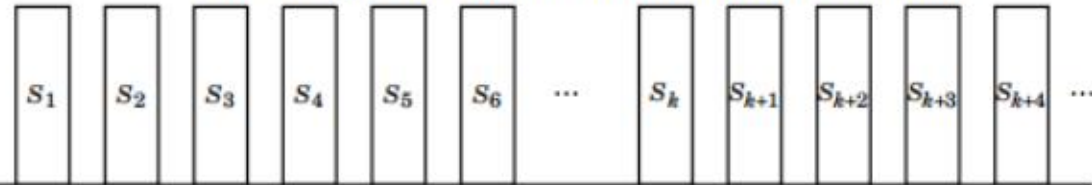


It looks good, however in the moment the correctness of our proof depends on whether there actually exists such k that the statement holds for it. The goal of the Base Case is exactly to show that there is a number for which the statement holds.

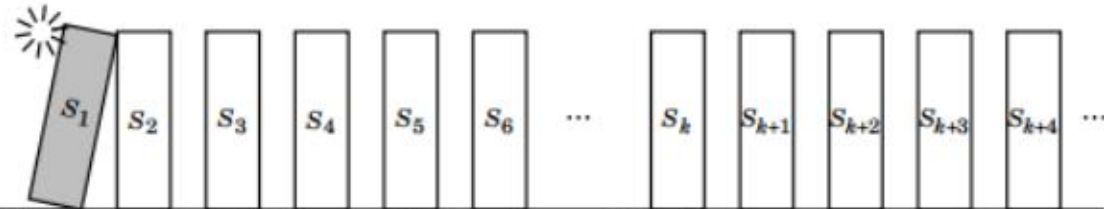
Strong induction

Strong induction has the same steps as the normal induction, however in the Induction Hypothesis instead of assuming that the statement holds for some $k \in \mathbb{N}$, we assume that the statement holds for all numbers from 1 to k included.

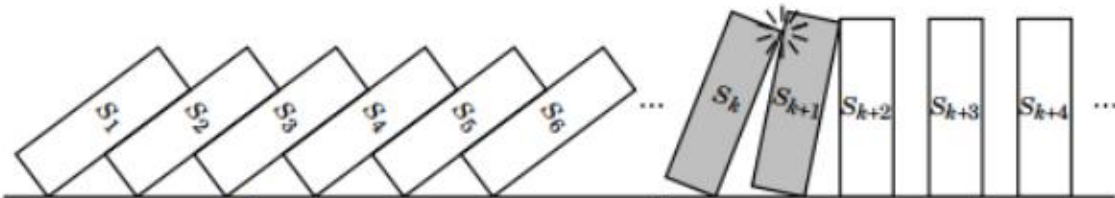
The Simple Idea Behind Mathematical Induction



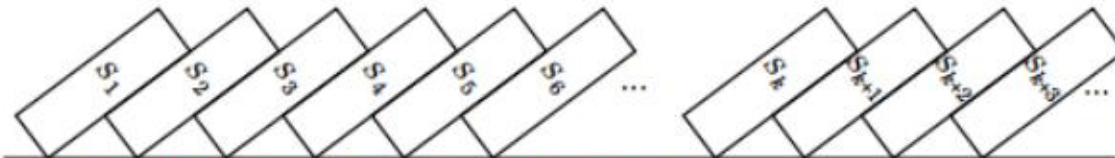
Statements are lined up like dominoes.



(1) Suppose the first statement falls (i.e. is proved true);



(2) Suppose the k^{th} falling always causes the $(k+1)^{th}$ to fall;



Then all must fall (i.e. all statements are proved true).

In Class Exercises

Limit of a Function

Informal Explanation A limit is the value that a function "approaches" as the argument gets closer and closer to a certain point. It describes the behaviour of the function when the argument gets near that point, rather than the function's value at the point itself.

$\lim_{x \rightarrow y} f(x)$, where f is the function, x is the argument, y is the point to which the argument goes closer and closer to.

Useful Rules Let $f(x), g(x)$ be two functions, such that
 $\lim_{x \rightarrow y} f(x) = c$ $\lim_{x \rightarrow y} g(x) = d$, where $c, d \in \mathbb{R}$ (this means $c, d \neq \infty$)

8/14

$$1) \lim_{x \rightarrow y} f(x) + g(x) = \lim_{x \rightarrow y} f(x) + \lim_{x \rightarrow y} g(x) = c + d$$

$$2) \lim_{x \rightarrow y} f(x) - g(x) = \lim_{x \rightarrow y} f(x) - \lim_{x \rightarrow y} g(x) = c - d$$

$$3) \lim_{x \rightarrow y} f(x) \cdot g(x) = \lim_{x \rightarrow y} f(x) \cdot \lim_{x \rightarrow y} g(x) = c \cdot d$$

$$4) \text{ if } \lim_{x \rightarrow y} g(x) = d \neq 0$$

$$\lim_{x \rightarrow y} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow y} f(x)}{\lim_{x \rightarrow y} g(x)} = \frac{c}{d}$$

Asymptotic Growth

When we estimate the number of elementary operations executed by algorithms, it is often useful to ignore smaller order terms, and instead focus on the asymptotic growth defined below. We denote by $\mathbb{R}^+$ the set of all (strictly) positive real numbers and by $\mathbb{R}_0^+$ the set of nonnegative real numbers.

Definition 1. Let $f, g : \mathbb{N} \rightarrow \mathbb{R}^+$ be two functions. We say that f grows asymptotically faster than g if $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$.

This definition is also valid for functions defined on $\mathbb{R}^+$ instead of $\mathbb{N}$. In general, $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)}$ is the same as $\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)}$ if the second limit exists.

Good to know

$$1) \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0 \Rightarrow n \text{ grows asymptotically faster than } \log n$$

$$2) \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0 \Rightarrow n^n \text{ grows asymptotically faster than } \log n$$

$$3) \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad 4) \lim_{n \rightarrow \infty} q^n = 0 \text{ for } q \in \mathbb{R} \text{ and } |q| < 1$$

$$5) \lim_{n \rightarrow \infty} \frac{n^c}{k^n} = 0, \text{ where } c \in \mathbb{R}, k \in \mathbb{R}^+ \text{ and } |k| > 1 \text{ are constants}$$

Theorem 1 (L'Hôpital's rule). Assume that functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ and $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are differentiable, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$ and for all $x \in \mathbb{R}^+$, $g'(x) \neq 0$. If $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = C \in \mathbb{R}_0^+$ or $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \infty$, then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}.$$

Useful Rules

Let f and g be differentiable functions

$$1) (f(x) + g(x))' = f'(x) + g'(x)$$

$$2) (f(x) - g(x))' = f'(x) - g'(x)$$

$$3) (f(x) \cdot g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$4) \left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x)g(x) - g'(x)f(x)}{g^2(x)}, \text{ when } g(x) \neq 0$$

$$5) (c \cdot f(x))' = c \cdot f'(x), \text{ where } c \in \mathbb{R} \text{ is a constant}$$

$$6) (f(x) + c)' = f'(x), \text{ where } c \in \mathbb{R} \text{ is a constant}$$

Table of derivatives

function

first derivative

$$f(x) = c, c \in \mathbb{R}$$

$$f'(x) = 0$$

$$f(x) = x$$

$$f'(x) = 1$$

$$f(x) = x^n, n \in \mathbb{R}$$

$$f'(x) = n \cdot x^{n-1}$$

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$f(x) = \ln x$$

$$f'(x) = \frac{1}{x}$$

$$f(x) = \log_a(x), a \in \mathbb{R}^+$$

$$f'(x) = \frac{1}{x \ln(a)}$$

Thank you for your attention
and participation!